

PAPER<i>297 — UQFF Superluminal Hubble Expansion Ratio η</i>exp = 3.328 > 1

Session: 84

PAPER297 — UQFF Superluminal Hubble Expansion Ratio $\eta_{\text{exp}} = 3.328 > 1$

First UQFF Module Where $v_{\text{exp}}/c > 1$ at System Boundary

Session: 84 **Module:** UNIVERSE_DIAMETER_UQFF_MODULE.cpp (26th C++ UQFF module — Observable Universe as System) **Copyright:** Daniel T. Murphy, March 17, 2026 **Classification:** Unique Physics — First UQFF Superluminal Expansion Parameter ($\eta_{\text{exp}} > 1$)

Abstract

The UQFF Observable Universe Diameter Module is the **first UQFF module** where the system boundary recession velocity exceeds the speed of light: $v_{\text{exp}} = H \times r_{\text{obs}} = 9.984 \times 10^8 \text{ m/s} = 3.328c$. The dimensionless Superluminal Expansion Ratio $\eta_{\text{exp}} = v_{\text{exp}}/c = 3.328 > 1$ is a new UQFF parameter encoding the cosmological property that the observable universe spans **3.328 Hubble lengths** ($r_{\text{obs}} = 3.328 \times r_{\text{H}}$). The Hubble-expansion coupling factor at t_{Hubble} is $(1 + H \times t_{\text{H}}) = 1.988$ — a near-doubling of the Newtonian base over cosmic time. All 25 prior UQFF modules had $\eta_{\text{exp}} < 1$ (sub-luminal expansion).

1. Physical Setup

System: Observable Universe **Observable universe radius:** $r_{\text{obs}} = 4.4 \times 10^{26} \text{ m}$ **Hubble constant:** $H = 70 \text{ km/s/Mpc} = 2.269 \times 10^{-18} \text{ s}^{-1}$ **Speed of light:** $c = 3.0 \times 10^8 \text{ m/s}$ **Hubble sphere (Hubble radius):** $r_{\text{H}} = c/H = 1.322 \times 10^{26} \text{ m} = 4.28 \text{ Gly}$ **Cosmic age:** $t_{\text{H}} = 4.355 \times 10^{17} \text{ s} = 13.8 \text{ Gyr}$

2. Master Relation

Hubble recession velocity at observable universe boundary:

$$v_{\text{exp}} = H_0 \times r_{\text{obs}} = 2.269 \times 10^{-18} \times 4.4 \times 10^{26} = 9.984 \times 10^8 \text{ m/s}$$

Superluminal Hubble Expansion Ratio:

$$\boxed{\eta_{\text{exp}} = \frac{v_{\text{exp}}}{c} = \frac{9.984 \times 10^8}{3.0 \times 10^8} = 3.328 > 1}$$

Hubble length ratio:

$$\frac{r_{\text{obs}}}{r_{\text{H}}} = \frac{v_{\text{exp}}}{c} = \eta_{\text{exp}} = 3.328$$

The observable universe boundary is 3.328 Hubble lengths from Earth — comfortably beyond the Hubble sphere where recession velocity equals c .

3. Hubble-UQFF Near-Doubling of Gravity

The base gravity term with Hubble expansion coupling:

$$a_{\text{base}}(t) = g_{\text{base}} \times (1 + H(z) \times t) = 3.447 \times 10^{-10} \times (1 + 2.269 \times 10^{-18} \times t)$$

At $t = t_H = 4.355 \times 10^8$ s:

$$a_{\text{base}}(t_H) = 3.447 \times 10^{-10} \times (1 + 0.988) = 3.447 \times 10^{-10} \times 1.988 = 6.854 \times 10^{-10} \text{ m/s}^2$$

Hubble expansion factor $\xi_H = 1 + H \times t_H = 1.988$:

This near-doubling factor (≈ 2) reflects that the UQFF base gravity **almost doubles** over the Hubble time when the Hubble coupling is included — a striking result confirming that the Universe-scale Hubble coupling is an $O(1)$ effect (not a small correction).

4. Superluminal Expansion in the EM Lorentz Term

The EM Lorentz term explicitly incorporates η_{exp} :

$$a_{\text{EM}} = \frac{q \cdot v_{\text{exp}} \cdot B_{\text{cosmic}}}{m_p} \times (1 + \eta_{\text{exp}}) \times \text{scale}_{\text{EM}}$$

With $\eta_{\text{exp}} = 3.328$, the factor $(1 + \eta_{\text{exp}}) = 4.328$ vs. $(1 + 1) = 2$ for a subluminal system. This introduces a **42% EM enhancement** relative to a hypothetical c-speed boundary reference.

Numerically:

$$\begin{aligned} a_{\text{EM}} &= \frac{1.602 \times 10^{-19} \times 9.984 \times 10^8 \times 10^{-15}}{1.673 \times 10^{-27}} \times 4.328 \times 10^{-12} \\ &= 95.59 \times 4.328 \times 10^{-12} = 4.136 \times 10^{-10} \text{ m/s}^2 \end{aligned}$$

The EM term ($4.136 \times 10^{-10} \text{ m/s}^2$) is **comparable to the Newtonian base** ($3.447 \times 10^{-10} \text{ m/s}^2$) — another first for UQFF modules.

5. Special Relativity Compatibility

The superluminal recession velocity $v_{\text{exp}} > c$ does NOT violate special relativity. This is a **coordinate velocity** (metric expansion), not a proper velocity between local inertial frames. In GR, the expansion of the universe allows coordinate distances to grow faster than c , as confirmed by the cosmological horizon structure. The Hubble-flow velocity $\eta_{\text{exp}} > 1$ is part of the cosmological metric, not a violation of local Lorentz invariance.

Specifically, this corresponds to objects beyond the **Hubble sphere** ($r > r_H$) in *comoving coordinates*. For the observable universe at $r = 4.4 \times 10^2 \text{ Mpc}$ with $r_H = 1.322 \times 10^2 \text{ Mpc}$, we are 3.33 Hubble lengths out — solidly in the superluminal expansion regime.

6. η_{exp} Parameter in UQFF Architecture

Module	Session	r_{obs} (m)	v_{exp}/c (η_{exp})	$\eta_{\text{exp}} > 1$
SGR1745	65	$\sim 0.01 \text{ pc}$	~ 0 (local)	No
Pillars of Creation	68	$3.3 \times 10^4 \text{ pc}$	$\gg 1$	No

Module	Session	r_obs (m)	vexp/c (ηexp)	η_exp > 1
NGC1792	73	7.6×10^{20}	■1	No
Andromeda	75	1.04×10^{21}	■1	No
HUDF (z=3.5)	72g	1.23×10^2 ■	~9 (co-moving)	Yes (but not UQFF param)
Universe Diameter	84	4.4×10^2■	3.328	Yes — FIRST explicit

The key distinction: prior modules used $H(z)$ as a correction factor, never explicitly computing or naming η_{exp} as a parameter. PAPER297 establishes η_{exp} as a first-class UQFF parameter.

7. Hubble Sphere as UQFF Speed-of-Light Boundary

Define the **UQFF Hubble Horizon** as the radius where $\eta_{\text{exp}} = 1$:

$$r_H = \frac{c}{H_0} = \frac{3 \times 10^8}{2.269 \times 10^{-18}} = 1.322 \times 10^{26} \text{ m} = 4.28 \text{ Gly}$$

Objects beyond r_H recede superluminally. The observable universe ($r_{\text{obs}} = 4.4 \times 10^2$ ■ m) extends to $3.328 r_H$ — confirming we can observe objects that are currently receding faster than light (their photons from early epochs can still reach us).

This adds a new entry to the UQFF speed-of-light boundary catalog:

- PAPER_264**: Anti-gravity boundary at $f_{\text{TRZ}} = -1$ (HUDF module)
- PAPER_266**: Meissner quench at $B = B_{\text{crit}}$
- PAPER_297**: Hubble superluminal horizon at $r = r_H$ (**this paper**)

8. WOLFRAM Term

```
eta_exp=v_exp/c=H0*r/c=9.984e8/3e8=3.328>1;
FIRST UQFF eta_exp>1;
r_H=c/H0=1.322e26m Hubble sphere;
r_obs=3.328*r_H;
expansion_factor(t_H)=1+H*t=1.988(near-doubling);
EM term a_EM prop eta_exp [PAPER_297]
```

9. Key Values Summary

Quantity	Symbol	Value	Unit
Hubble recession velocity	v_exp	9.984×10■	m/s
Superluminal ratio	η_exp	$3.328 > 1$	dimensionless
Hubble radius	r_H	1.322×10^2 ■	m
Observable / Hubble ratio	r_{obs}/r_H	3.328	dimensionless
Hubble expansion factor	ξ_H	$1.988 \approx 2$	dimensionless
EM term	a_EM	4.136×10^{-10}	m/s ²

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \sqrt{SSq}/\sqrt{GM/r} = 5.0e-4 \sqrt{0.57 \sqrt{6.67e-11} M/r}$; for solar parameters: $U_{bi,Sun} = 5.7e-4 \sqrt{6.67e-11} \sqrt{1.99e30/(6.96e8)} = 1.47e+2 \text{ m/s}$.